muscle movements, enabling a computer program to figure out what he was trying to say and then translate his Mandarin into English. The result boomed out of a loudspeaker a few seconds later:

"Let me introduce our new prototype," a synthesised voice announced. "You can speak in Mandarin and it translates into English or Spanish."

This might sound like we've reached what's needed in MT, but the fact is that electrodes, and the fact that speech is directly being translated into speech on the fly, do not make for a better translator! The article in the Post-Gazette goes on to say that when (he) announced he would take questions from reporters in Germany and America, the computer heard it as "so we glycogen it alternating questions between Germany and America." And, of course, things such as humour would be entirely lost in translation, to use the old phrase.

What's interesting about this system—of which several versions are being developed, by the way—is that it uses the statistical approach we mentioned earlier: a system learning from a body of texts in both languages, and how the translation was done between those texts. We can't stress enough that the statistical approach seems to us the most likely way to go for the future. And not just for the academic reason, but also for the fact that the mass of information (in all languages) on the Internet will increase with time, so machine translators will have that much more wisdom to draw from.

MT In India

It's probably in the EU and in India that the need for MT most exists. We came across two Indian projects of note—the Tamil-Hindi Machine Aided Translation System, at www.au-kbc.org/research_areas/nlp/demo/mat/ , and "An English to Hindi Machine Aided Translation System: an ongoing project at IIT Kanpur," at http://anglahindi.iitk.ac.in/ .

It's probably in the EU and in India that the need for Machine Translation most exists

According to a document on the Anglabharti project—which aims at translating English into all major Indian languages, and of which Anglahindi is a subset—the system uses something like an interlingua approach (the one that converts sentences from the source language into an intermediate system). It analyses English sentences to create an intermediate structure with most of the disambiguation performed in this step. The intermediate language structure has the word and word-group order according to the structure of the group of target (Indian) languages. Note that word order is an important consideration, since English usually strictly follows the subject-verb-object structure, whereas Indian languages follow a relatively free structure. As an example, consider

I am going to the shop.

This sentence, like most simple English sentences, has the subject ("I") first, then the verb ("going"), then the object ("the shop"). Now consider that it can be translated into Hindi as

Main dukaan ja raha hoon.
Dukaan ja raha hoon main.
Ja raha hoon main dukaan.
Main ja raha hoon dukaan.

Although one or two of these variants are more likely to be used, the fact is they're all grammatically correct. In contrast, there's no way of rephrasing the English sentence.

Therefore, converting the intermediate structure to each Indian language through a process of text generation is a relatively simple process. In fact, the authors of the paper say that the effort involved in analysing the English is about 70 per cent of all the work, while the text generation accounts for only 30 per cent. Part of the reason for this is the word order situation we've described.

If you speak Tamil and/or Hindi, try these out, and do remember to write in—we're especially interested in the bloopers!

A Universal Grammar?

A possible future direction for MT would be to take advantage of a linguistics theory called "Universal Grammar" (UG). From Wikipedia, UG postulates principles of grammar shared by all languages, thought to be innate to humans. It attempts to explain language acquisition in general, not describe specific languages. UG proposes a set of rules that would explain how children acquire their language.

According to UG, there is *only one method* of "diagramming" sentences; this method applies to all the languages of the world, and is universal because it is genetically encoded into the brain of every human child. This is a bold thesis, and a large number of linguists are working within this approach.

MT would benefit from a well-developed UG theory because the intermediate-language approach could easily be used to generate a representation of the meaning of the sentence, since there is only one way of diagramming it. Problems such as those of ambiguity resolution would remain, but the problems of syntactic and semantic parsing would be solved.

We Hate To Tell You, But...

MT is still in its infancy. Worse, it's been in its infancy since the 50s. Each researcher proposes a different scheme for "the MT system of the future," based on one or the other of the basic approaches. Some say we will *never* have completely automated MT, and sadly, that just could be true. In the foreseeable future, MT seems best focused at aiding human translators.

The most promising thing we can tell you is that the statistical approach based on large bodies of existing translation can only get better as the information on the WWW increases: we're putting our bets on Google, by the way! Which, reverse translated to and from German—even if that's unfair—becomes "We set our bets on Google, by the way!" And that's amazing... there might still be hope! ■

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